

MOHAMMAD TANBERUL ISLAM

Electrical & Electronics Engineer

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🌐 circuitsecrets.blogspot.com | 📍 Chittagong, Bangladesh

PROFESSIONAL SUMMARY

Results-driven Electrical & Electronics Engineer with 6+ years of progressive industrial experience in electrical maintenance, preventive maintenance planning, and ERP-based operations management within a large-scale garment manufacturing facility. Holds a BSc in Electrical & Electronics Engineering (CGPA: 3.59) complemented by academic research in Field-Oriented Control (FOC) of induction motors using MATLAB/Simulink. Proven track record of reducing equipment downtime, strengthening inventory control systems, and improving facility-wide electrical reliability. Seeking a challenging technical engineering role in Electrical Power Systems, Industrial Automation, or Control Systems — with a strong growth trajectory toward PLC/SCADA operations and energy efficiency management.

TECHNICAL SKILLS

Electrical Systems & Maintenance

Electrical Maintenance & Troubleshooting • Induction Motor Systems • Preventive & Breakdown Maintenance • AC/HVAC Systems • Industrial Wiring & Installation • Electrical Safety Standards

Simulation & Design Tools

MATLAB / Simulink • AutoCAD Electrical (Fundamentals)

Operations & Management Systems

ERP-Based Inventory Management • Maintenance Documentation • Procurement Coordination • Audit-Ready Technical Reporting

Control & Automation (Foundational)

PLC Concepts & Control Fundamentals • Field-Oriented Control (FOC) • Motor Drive Theory • Feedback Control Systems • Energy Efficiency Optimization

Power & Electrical Fundamentals

Three-Phase Power Systems • Motor Protection & Control Circuits • Transformer Fundamentals • Refrigeration & Air Conditioning (RAC)

Software & Productivity

Microsoft Word / Excel / PowerPoint • Technical Blog Writing • Email & Digital Communication

PROFESSIONAL EXPERIENCE

Assistant Executive — Engineering

March 2019 – Present

Four H Fashion Ltd (Four H Group) | Chittagong, Bangladesh

- Managed ERP-based central electrical store operations, maintaining real-time inventory accuracy for 500+ line items — eliminating procurement delays and reducing unplanned material shortfalls by an estimated 30%.
- Designed and implemented a structured preventive maintenance (PM) schedule covering electrical systems, AC units, and industrial motors — resulting in measurably reduced emergency breakdown incidents and improved overall equipment effectiveness (OEE).
- Led rapid-response electrical troubleshooting across production and utility zones, consistently minimizing unplanned downtime and sustaining compliance with electrical safety standards.
- Coordinated multi-department maintenance workflows across Engineering, Operations, and Procurement teams to ensure seamless, facility-wide electrical infrastructure reliability.
- Maintained comprehensive maintenance records, equipment history logs, and audit-ready technical documentation — supporting internal audits, regulatory compliance, and capital planning decisions.
- Supervised performance monitoring of AC systems and industrial motor arrays, identifying early failure indicators and executing proactive interventions to prevent production stoppages.

Key Impact: Achieved sustained improvement in equipment uptime and operational reliability; reduced emergency maintenance incidents through systematic PM scheduling; strengthened real-time inventory accuracy via full ERP integration.

ACADEMIC BACKGROUND

Degree / Qualification	Institution	Year	Grade
BSc – Electrical & Electronics Engineering (EEE)	Southern University Bangladesh	2025	CGPA: 3.59
Diploma in Electrical Engineering	Chittagong Polytechnic Institute (BTEB)	2018	CGPA: 3.06
SSC – Science	Khaiyachara High School, Chattogram Board	2014	GPA: 4.75

ACADEMIC THESIS / RESEARCH

Efficiency Improvement of an Induction Motor Using Advanced Control Techniques *BSc Final Year | 2025*

Southern University Bangladesh | MATLAB / Simulink | Field-Oriented Control (FOC)

Conducted an in-depth research project targeting energy efficiency improvement in three-phase induction motors through advanced closed-loop control strategies. The work addressed limitations of conventional scalar (V/f) control and investigated higher-performance alternatives relevant to variable-speed industrial drive applications.

- Designed and implemented a Field-Oriented Control (FOC) model in MATLAB/Simulink, achieving decoupled control of motor flux and torque for precise speed regulation and enhanced dynamic response.
- Developed a complete induction motor mathematical model in Simulink, incorporating stator and rotor parameters, magnetizing inductance, and speed-dependent torque characteristics for simulation accuracy.
- Conducted performance analysis under varying speed and load transient conditions — validating that FOC implementation reduced torque ripple and improved efficiency over open-loop and scalar control baselines.
- Evaluated energy efficiency metrics across multiple operating points, demonstrating measurable reductions in copper and iron losses at partial-load — directly applicable to industrial energy conservation use cases.
- Produced a structured technical thesis report with detailed block diagrams, simulation schematics, and comparative performance data.

Technical Keywords: Field-Oriented Control (FOC) • Induction Motor Modelling • MATLAB/Simulink • Variable Speed Drive (VSD) • Torque-Flux Decoupling • Energy Efficiency • Motor Control Systems

TRAINING & CERTIFICATIONS

Training / Certification	Institution	Location	Year / Duration
City & Guilds London Level 2 – Electrical & Electronics Engineering	BKTTC	Nasirabad, Chittagong	2018 6 Months
Electrical Training (STEP Programme)	BKTTC	Nasirabad, Chittagong	2018 6 Months
Industrial Training – Electrical Systems	Eastern Cables Ltd.	Mohazangata, Chittagong	2018 3 Months
RAC Level 1 – Refrigeration & Air Conditioning (SEIP)	BKTTC	Nasirabad, Chittagong	2016 6 Months

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Technical Blogger — CircuitSecrets

2024 – Present

circuitsecrets.blogspot.com

- Author and publish technical tutorials and practical insights on electrical and electronics engineering — supporting knowledge sharing for students and early-career professionals globally.
- Demonstrate strong technical writing skills, subject-matter authority, and commitment to continuous learning through consistent content creation and platform maintenance.

CORE COMPETENCIES

Analytical Problem-Solving • Cross-Functional Team Collaboration • Technical Documentation & Reporting
• Proactive Maintenance Planning • Time & Priority Management • Clear Engineering Communication • Continuous Self-Development

LANGUAGES

English: Professional Working Proficiency **Bangla:** Native / Full Professional Proficiency

REFERENCES

Prof. Ashutosh Nath

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I hereby declare that all information provided is true and accurate to the best of my knowledge. Supporting documents available upon request.

MOHAMMAD TANBERUL ISLAM